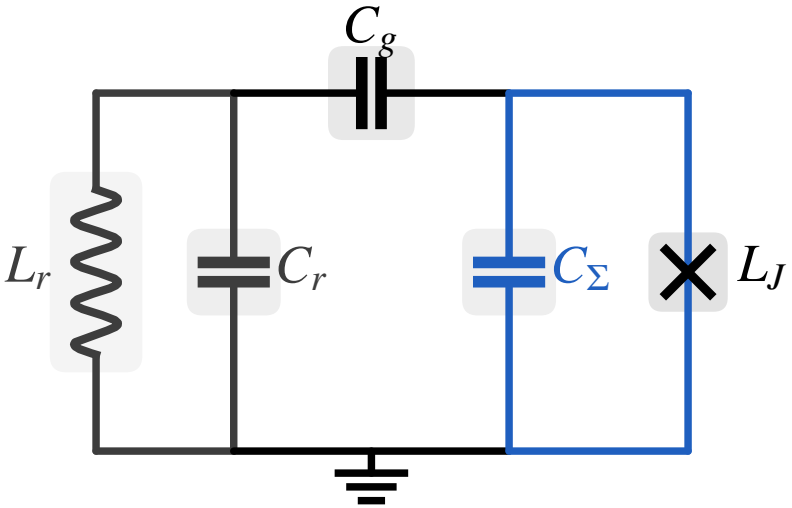
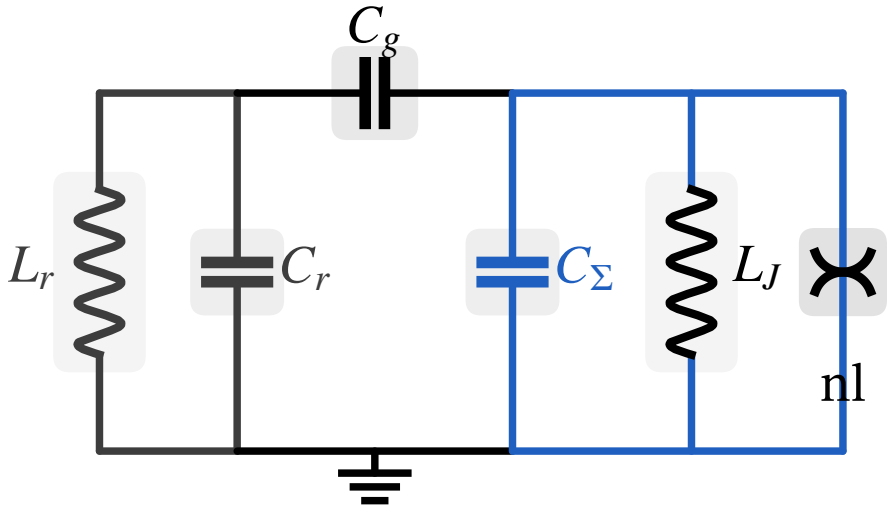
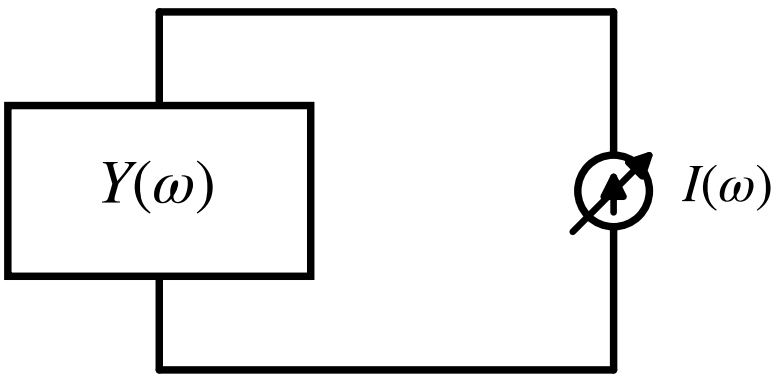
(a)



(b)

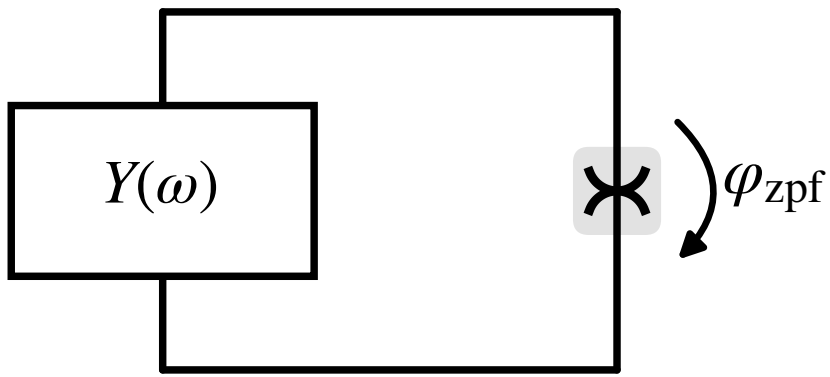


(c)



$Y(\omega) = I/V$

(d)



$\varphi_q = 0.41, \varphi_r = 0.026$